

Week 4 Progress Report: Adaptive Item Selection Results

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Date: August 10 - August 19, 2026

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Summary

Following your go-ahead to implement the paper's actual item selection approach, I built and ran BanditCAT V1 style adaptive item selection (Thompson sampling on Fisher information, Gamma-randomized exposure control, partial random injection for coverage) in place of the uniform random draw used in every DET run so far. The headline result is the opposite of what I expected going in: adaptive selection underperforms random selection for item calibration at every condition tested, at both a small item bank and the real DET scale. I checked whether this was a bug, checked whether it was the correlation-inflation artifact the AutoIRT paper itself warns about, and traced the actual mechanism. As part of that, I also pressure-tested whether the full-scale gap was sensitive to total response volume, since adaptive selection's exposure is non-uniform in a way random selection's isn't; it was, and controlling for it brings the full-scale gap down to roughly the same size as the small-scale gap, rather than the much larger gap in an earlier pass. Details, numbers, and an interpretation follow.

1. Headline finding

At both scales tested, item-grade Pearson correlation is lower under adaptive selection than under random selection, for every split condition and both item types. At the real DET scale, I also tested whether the size of the gap was sensitive to total response volume: it is. Matching the sessions-per-item ratio used in the small-scale check (rather than just a larger absolute session count) closes much of the gap, and the remaining, smaller gap is fairly consistent with what the small-scale results already showed. Section 3 has the details.

2. Small-scale results (150 Y/N Vocab / 80 ViC items, 6,000 sessions, 10 replications each)

Item-grade Pearson r , mean over 10 replications.

Y/N Vocab

Split	Random	Adaptive	Difference
Cold	0.619	0.396	-0.223
Jump 20	0.857	0.674	-0.183
Jump 40	0.898	0.759	-0.139
Jump 80	0.928	0.803	-0.125
Warm 05-08	0.965	0.899	-0.066
Warm 05-15	0.965	0.901	-0.064

ViC

Split	Random	Adaptive	Difference
Cold	0.572	0.463	-0.109

Jump 20	0.903	0.796	-0.107
Jump 40	0.927	0.859	-0.068
Jump 80	0.943	0.880	-0.063
Warm 05-08	0.963	0.941	-0.022
Warm 05-15	0.964	0.944	-0.020

Every condition drops under adaptive selection. The drop is largest for Cold and shrinks, but never disappears, moving toward Warm-start, likely because warm-start's larger training set partially compensates.

3. Full-scale results (3,290 Y/N Vocab / 585 ViC items, real DET item bank size)

Random uses config.*N_DET_SESSIONS* = 25,000 sessions, averaged over 3 replications. Adaptive selection is non-uniform, so it needs more total response volume than random selection to reach the same depth per item; the first full-scale adaptive pass used *N_DET_SESSIONS_ADAPTIVE* = 50,000, which turned out to still be a noticeably thinner sessions-per-item ratio than the small-scale check had used (15.2 vs. 40.0 sessions per item). Raising *N_DET_SESSIONS_ADAPTIVE* to 131,600 to match that ratio closes most of the gap seen at 50,000, shown below. This confirms that a meaningful part of the originally observed full-scale gap was a response-volume effect layered on top of the restricted-range effect, not the restricted-range effect alone. The adaptive column below is a single run at 131,600 sessions rather than a 3-replication mean, so it should be read as a solid first read rather than as tight as the small-scale and random-mode numbers.

Y/N Vocab

Split	Random	Adaptive	Difference
Cold	0.758	0.574	-0.184
Jump 20	0.798	0.675	-0.123
Jump 40	0.717	0.665	-0.052
Jump 80	0.858	0.706	-0.152
Warm 05-08	0.847	0.618	-0.229
Warm 05-15	0.847	0.600	-0.247

ViC

Split	Random	Adaptive	Difference
Cold	0.661	0.589	-0.072
Jump 20	0.852	0.743	-0.109
Jump 40	0.889	0.771	-0.118
Jump 80	0.907	0.815	-0.092
Warm 05-08	0.926	0.871	-0.055
Warm 05-15	0.929	0.869	-0.060

At the corrected session volume, the full-scale gap (roughly 0.05 to 0.25 Pearson) is much closer in size to the small-scale gap (roughly 0.02 to 0.22) than the 50,000-session run suggested. One notable shift: Warm-start now shows the largest gap of the six conditions for Y/N Vocab (-0.229, -0.247), rather than the smallest. Warm-start has no pilot items and no R-depth requirement, so it is not subject to the same response-volume

confound that affects Cold and Jump-start, which makes it arguably the cleanest single read on the restricted-range mechanism specifically.

4. Ruling out the correlation-inflation artifact

The AutoIRT paper's own "Evaluating Calibrated Item Parameters" section warns that non-random administration can inflate item-grade correlation even when item parameters are constant, because the ability estimate used to compute predicted grades is itself derived from the same non-randomly-selected responses. If that artifact were driving these results, adaptive selection should look artificially better on Pearson while test loss stays flat or worsens only slightly.

That is not what happened at either session count. Mean test loss (binary cross-entropy) is also worse under adaptive selection at full scale: at the corrected 131,600-session run, Y/N Vocab moved from about 0.55 (random) to about 0.59-0.63 (adaptive); ViC moved from about 0.50-0.55 (random) to about 0.56-0.65 (adaptive). Both metrics move in the same, worse direction together at both session counts tested, which is the signature check_adaptive_selection.py's own documentation predicted for a genuine restricted-range problem, and the opposite of what the correlation-inflation artifact would produce. I am treating this as a real, if now more moderate, calibration effect, not a measurement illusion.

5. Mechanism: restricted per-item ability range

check_adaptive_selection.py, written before this work started, flagged the likely cause in advance: adaptive selection concentrates each item's exposures near test-takers whose ability is close to that item's own difficulty, by design, since that is what makes it efficient for scoring. Naive batch calibration needs the opposite: each item needs to be seen across a wide range of abilities to fit a reliable difficulty/discrimination curve. DET_ADAPTIVE_RANDOM_INJECTION_RATE = 0.4 (40 percent of item-selection rounds fall back to picking uniformly at random) was chosen to partially restore this coverage, and the diagnostic script's own coverage sweep showed it helping at a smaller check scale.

Matching the small-scale check's sessions-per-item ratio (131,600 sessions rather than 50,000) closed most of the full-scale gap, which means a good part of what looked like a severe restricted-range effect at 50,000 sessions was actually a response-volume shortfall on top of it. A real, smaller gap remains after controlling for that, and Warm-start is the cleanest place to see it: Warm-start has no pilot items and no R-depth requirement, so it is not affected by the same volume confound Cold and Jump-start were, and it still shows the largest gap of the six conditions for Y/N Vocab. That is consistent with restricted ability-range coverage being the remaining driver, separate from data volume.

6. n_steps = 7 at full scale: revisited

In the 50,000-session run, every one of the 12 full-scale adaptive conditions converged at exactly 7 EM steps (standard deviation 0 across 3 replications). I had read that as the training loss being flat from essentially the first step, since step 7 is the earliest point autoirt_model.py's convergence check can ever declare convergence (it needs a window of 6 relative-loss-change flags, and those only start accumulating from the second EM step onward).

The 131,600-session run shows that reading was incomplete. Cold and Jump-start still stop at exactly step 7 even with far more data, but Pearson roughly doubled for those same conditions, and the per-step training loss in the log shows a steady decline across all 7 steps, not a flat line. Warm-start, which now has substantially more training data than before, runs 10 to 13 steps instead of stopping at 7, and keeps improving. So step 7 for

Cold and Jump-start looks less like the model running out of signal and more like the convergence check's per-step relative-change threshold being satisfied while real, gradual improvement is still available. That is a more encouraging read than the original one: it points to a tunable stopping-rule parameter for these two conditions specifically, rather than a hard ceiling, and is worth a follow-up on its own.

7. A second effect: rank order degrades more than the linear correlation does

At small scale, item-grade Spearman correlation drops more sharply than Pearson under adaptive selection. For example, Y/N Vocab Warm 05-15 drops from Pearson 0.901 mean to Spearman 0.582 mean under adaptive (a gap of 0.319), versus Pearson 0.965 to Spearman 0.904 under random (a gap of 0.061). The same pattern holds for Cold and for ViC. The 131,600-session full-scale run shows the same shape: Y/N Vocab Warm 05-15 is Pearson 0.600 but Spearman 0.364 (a gap of 0.236), and ViC Warm 05-15 is Pearson 0.869 but Spearman 0.495 (a gap of 0.374), both well beyond the corresponding random-mode gaps. Seeing the same pattern at both scales makes it more likely to be a real, reproducible effect rather than small-scale noise, but I still do not have a clean explanation for why relative item ranking is harmed more than the linear fit is, and I am leaving it as an open question rather than guessing at a cause.

8. A possible reframing

The AutoIRT paper's cold-start and jump-start use cases assume operational items already have a broad response history to train from. In the real DET, that history comes from however operational items were actually administered historically, which for launched items includes a meaningful stretch of non-adaptive or less aggressively adaptive exposure before BanditCAT-style selection took over (see the BanditCAT paper's own account of the phased rollout, E1 through E5, before the final launch). One reading of this result is that adaptive administration and item calibration are different objectives with a real tension between them: BanditCAT-style selection is good for test-taker scoring efficiency once items are already calibrated, but is a poor source of training data for calibrating those same items in the first place. That would argue for keeping calibration and administration data separate rather than calibrating directly from adaptively-selected responses, which is closer to how the papers describe the real system operating.

9. Caveats

- The random-mode full-scale numbers are a 3-replication mean at 25,000 sessions. The adaptive full-scale numbers at 131,600 sessions are a single run, not yet replicated. The direction and rough size of the effect line up well with the small-scale (10-replication) results, which gives some confidence in the single run, but the exact full-scale adaptive values should be treated as a solid first read rather than as tight as the other numbers in this report until replicated.
- Y/N Vocab and ViC Jump 20/40/80 produced clearly distinct training set sizes at 131,600 sessions (813,855 / 846,755 / 912,554 for Y/N Vocab), so the collapse-into-each-other failure mode from the original 1,500-session run is not happening here. I have not separately reconfirmed the pilot-item coverage percentage for Jump 80 at this session count the way I did for the 50,000-session run, and would want to before treating that row as fully settled.
- The Spearman-collapse finding in Section 7 is reported descriptively; no cause has been confirmed yet.

Next steps

- Replicate the 131,600-session full-scale adaptive run (2-3 seeds, matching the fullscale random-mode standard) to get a tighter estimate before treating the updated gap size as final.

- Look into the step-7 early-stop behavior for Cold and Jump-start specifically, since the corrected run suggests real, recoverable calibration gain may be left on the table there.
- My inclination otherwise is to treat the directional finding as settled (adaptive selection underperforms random for calibration at every condition tested, at both scales) and return focus to the jump-start ML/MCEM gap identified in Week 3. Let me know if you would like a different priority.